

2024 AMC 10B Problem 3 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10B Problem 3. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10B Problem 3

Problem:

For how many integer values of x is $|2x| \leq 7\pi$?

Answer Choices:

- A. 16
- B. 17
- C. 19
- D. 20
- E. 21

Solution:

From $3 < \pi < 3.142$, it follows that

$$21 = 7 \cdot 3 < 7\pi < 7 \cdot 3.142 = 21.994 < 22$$

Because x is an integer, the values for $2x$ that make the inequality true are $-20, -18, -16, \dots, -2, 0, 2, \dots, 18$, and 20 . Each of these corresponds to a unique value of x . There are (E) 21 such values.

The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) ↗

Powered by [Wiki.js](#)